

FIELD INTELLIGENCE OS

v1.0 Product Specification

Build-ready functional, technical, data, AI, integration, security, mobile, and operations specification

Product thesis

Field Intelligence OS is the primary application through which a field investigator plans deployments, manages people and institutions, captures multimodal evidence, reconstructs systems, verifies claims, preserves permissions, and produces publication-ready outputs. Notion and Google Drive are synchronized secondary layers: Notion is the human-readable knowledge mirror; Drive is the durable media and source-document vault. Supabase is the canonical application database. Vercel hosts the application.

Field	Value
Specification owner	RN Collins
Version	1.0
Status	Build-ready baseline
Prepared	19 July 2026
Primary users	RN Collins initially; later investigators, editors, researchers, educators, and teams
Primary stack	Next.js / TypeScript / Supabase / Vercel / Google Drive API / Notion API
Companion artifacts	schema.sql, openapi.yaml, screen inventory, permission matrix, event model, migration map

Document map

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How engineering should use this document

The numbered requirements are normative. Tables define required behavior unless explicitly labeled optional or future. Where a workflow, screen, API, and table disagree, the canonical data model and permission rules control; the product owner resolves remaining conflicts through a recorded architecture decision.

1. Executive product definition

1.1 Product mission

Create one operational environment in which fieldwork moves continuously from question to deployment, source relationship, consent, capture, evidence, verification, systems reconstruction, synthesis, publication, and longitudinal follow-up.

1.2 Product category

A multimodal field-intelligence operating system. It combines research operations, editorial source management, field capture, evidence management, knowledge graph functions, asset provenance, and output production. It is not merely a CRM, DAM, note-taking tool, project manager, or generative-writing application.

1.3 Canonical architecture

The application is the authoritative operational system. Supabase/Postgres holds canonical structured records. Supabase Storage may hold lightweight derived assets, thumbnails, transcripts, and app-generated files. Google Drive holds original media masters and source-document copies. Notion mirrors selected records and provides durable human-readable pages. Vercel hosts local, preview, and production environments. GitHub is the code and schema history.

1.4 Initial release objective

Deliver a secure single-user production system that can operate Manhattan and Reykjavík as live deployments, complete an end-to-end field investigation, ingest and link captured media, verify claims, and assemble outputs without returning to disconnected spreadsheets.

1.5 Success condition

RN can open one homepage and know: what must happen today; which interviews require action; which claims remain unverified; which capture packages are incomplete; which files await ingest; which promises are due; which outputs can be produced; and how every item connects to a person, institution, place, system, law, and permission record.

1.6 Release acceptance metrics

Metric	v1.0 acceptance threshold
Operational completeness	A test deployment can progress through target → outreach → booking → consent → capture → ingest → transcript → claim extraction → verification → system map → follow-up → output without leaving FIOS except for external communication and file capture.
Retrieval	A user can retrieve every asset, source, claim, law, interaction, and output related to a deployment, person, institution, system, or place within three interactions.
Provenance	Every publishable claim can be traced to one or more evidence records and every media asset to its capture context and permission status.
Resilience	No single failure of the application, Notion, or one local device causes loss of irreplaceable field media or canonical structured records.

Metric	v1.0 acceptance threshold
Field usability	The core mobile capture flow can be completed with one hand in no more than three taps after opening Field Mode.
Trust	AI-generated suggestions are never silently accepted as facts, quotations, permissions, legal conclusions, or editorial decisions.

2. Product principles and scope

Principle	Operational meaning
Reality before representation	The system prioritizes preserving irrecoverable real-world information before generating summaries or polished media.
One record, many uses	A person, interaction, claim, or asset is entered once and related to every relevant project or output.
Claim and evidence are separate	A statement is not evidence merely because it came from an expert, institution, slide, or company.
Permissions travel with the asset	Attribution, consent, embargo, privacy, and role boundaries remain attached to records and derivatives.
Human judgment remains explicit	The system identifies where human interpretation, override, responsibility, and uncertainty enter.
Place is data	Geography, architecture, infrastructure, climate, language, and movement are first-class entities.
Longitudinal by default	Records remain connected across years, trips, jurisdictions, and repeated encounters.
AI assists; humans decide	AI may classify, transcribe, retrieve, compare, and propose. It may not invent, finalize, publish, or change permissions autonomously.
Fail-safe, not duplicate labor	Notion and Drive mirrors exist for resilience and portability, but the user should not manually maintain three competing systems.
Progressive disclosure	Desktop may expose complexity; Field Mode reveals only the next necessary action.

2.1 In scope for v1.0

- Authentication and single-user workspace; architecture must be multi-tenant ready.
- Command Center; deployment workspaces; targets and relationships; interactions and interviews; tasks and commitments.
- Claims, evidence, law/policy, contradictions, silence/absence records, and verification queues.

- Systems reconstruction, places/routes, assets, permissions, cognition updates, and outputs.
- Mobile Field Mode, quick capture, offline draft queue, audio note intake, capture-package completion.
- Google Drive folder creation, file linking, metadata synchronization, and ingest status.
- Notion page/database mirroring with deterministic external IDs and conflict logs.
- AI transcription intake, extraction suggestions, semantic retrieval, brief generation, and output assembly support.
- Manhattan and Reykjavík seed data and full live workflows.
- Audit logs, version history, export, corrections, and role separation.

2.2 Explicitly out of scope for initial production

- Native camera replacement, professional non-linear video editing, or long-term storage of all original masters inside the app.
- Automatic publication to outlets without human review and explicit authorization.
- Autonomous outreach, interviewing, source consent, legal advice, or fact verification.
- Public multi-user marketplace, certification, billing, or open-standard governance.
- Full digital evidence chain-of-custody suitable for criminal litigation; v1.0 preserves strong provenance but is not a forensic evidence platform.
- Biometric identification, facial recognition, covert recording, or surveillance features.

3. Users, roles, and jobs to be done

Role	Initial availability	Primary job
Workspace Owner	RN initially	Create and administer workspace, integrations, permissions, standards, deployments, exports, and retention.
Investigator	Primary daily role	Plan, capture, interview, observe, verify, reconstruct systems, and create cognition updates.
Editor/Reviewer	Future collaborator	Review claims, evidence, drafts, corrections, permissions, and publication readiness without changing protected masters.
Research Assistant	Future collaborator	Research targets, laws, documents, and evidence; cannot publish or override consent.
Producer/Archivist	Future collaborator	Ingest, rename, back up, transcribe, tag, and prepare derivatives; cannot alter editorial conclusions.
Viewer/Client	Future restricted role	Read approved dashboards or deliverables; no access to source identities or restricted records unless granted.
Service Integration	Not human	Notion, Drive, transcription, calendar, email, and automation service accounts with least-privilege scopes.

3.1 Core jobs to be done

Situation	System job
When I am planning a trip	help me convert broad ambitions into a ranked, schedulable, evidence-seeking deployment.
When I identify a person or institution	show why they matter, what to request, what boundaries apply, and the precise next action.
When I am entering a venue	show only the route, question, permissions, and minimum capture package.
When a claim is made	let me preserve the exact wording, context, evidence, limits, counterpoint, and verification status.
When an interview ends	help me protect the file, capture what changed, log promises, and identify missing reality.
When media arrives	create a reliable ingest trail and link each asset to its context without requiring repetitive manual data entry.
When writing or editing later	retrieve the complete evidentiary sequence rather than a pile of clips and notes.
When a source restriction changes	propagate the restriction to connected assets and outputs and alert me to affected work.
When I revisit a place years later	show prior people, laws, claims, routes, systems, unanswered questions, and changes over time.

4. System context and architecture

4.1 Logical context

User devices (desktop / mobile / tablet)

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Next.js web application + installable PWA on Vercel

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+--> Supabase Auth

+--> Postgres canonical records + Row Level Security

+--> Supabase Storage for thumbnails, transcripts, derivatives

+--> Edge Functions / background job API

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+--> Google Drive API: original media and source-document vault

+--> Notion API: curated knowledge mirror and portable human interface

+--> AI providers: transcription, embeddings, extraction, retrieval, generation

+--> Optional Gmail / Calendar: outreach and interview operations

+--> Observability: application logs, errors, audit events, job telemetry

Component	Recommended implementation	Responsibility
Web client	Next.js App Router, TypeScript, responsive	Primary user interface, field mode,

Component	Recommended implementation	Responsibility
	PWA	dashboard, local queue, upload/link flows.
Application API	Next.js route handlers and/or Supabase Edge Functions	Authorization-aware business logic, integrations, job triggers, exports.
Canonical database	Supabase Postgres	All structured operational records, relationships, external IDs, audit and sync state.
App file layer	Supabase Storage	Thumbnails, transcripts, low-resolution proxies, generated briefs, export packages; not default location for original large media.
Media/source vault	Google Drive	Original camera/audio/photo files, source documents, consent files, immutable master organization.
Knowledge mirror	Notion	Curated deployment pages, readable source pages, dashboards and manual/standards mirror.
Search/RAG	Postgres full-text + pgvector; optional external reranker	Keyword, filter, semantic, graph-neighborhood, and hybrid retrieval.
Job system	Postgres jobs table + Edge Function/queue provider	Transcription, embeddings, Drive sync, Notion sync, extraction, exports, checksums.
Hosting	Vercel local/preview/production	Application deployment and preview workflow; Git-connected continuous deployment.
Source control	GitHub	Code, migrations, schema, tests, ADRs, issue/PR workflow.

4.2 Source-of-truth rules

Information	Canonical location	Rule
Structured entities and state	Supabase/Postgres	Notion edits are imported only through explicit sync rules; no silent overwrite.
Original media masters	Google Drive	Database stores Drive file ID, metadata, hashes when available, and permissions.
Derived transcripts and proxies	Supabase Storage + Drive copy where required	Database stores canonical current version and lineage.
Human-readable mirrored pages	Notion	Regenerable from canonical database; Notion-only commentary may be imported as interaction notes.

Information	Canonical location	Rule
Code and schema	GitHub	Production changes require versioned migrations and review.
Published outputs	Outlet/platform plus archived final in Drive	Database stores publication URL, archive file, rights, corrections, and source set.

4.3 Environment model

Environment	Purpose	Data	Integrations	Access
Local	Development and unit testing	Synthetic/local	Sandbox or disabled	Developer only
Preview	Feature QA for each branch/PR	Seeded non-sensitive test workspace	Test integrations; never production secrets where avoidable	Authorized reviewers
Staging	Pre-production workflow and migration tests	Sanitized copy or dedicated staging data	Dedicated Notion/Drive test folders	Owner + developers
Production	Live field operations	Canonical live data	Production integrations and vault	Workspace users

5. Information architecture and navigation

5.1 Global navigation

Navigation item	Purpose
Command Center	Current state, urgent queues, active deployments, quick capture.
Deployments	Trips, events, residencies, assignments, daily plans and fieldbooks.
People & Institutions	Targets, relationships, organizations, roles, expertise and history.
Interactions	Outreach, interviews, meetings, panels, observations and correspondence.
Claims & Evidence	Claims, evidence, law/policy, contradictions, absences and verification.
Systems & Places	Workflow reconstructions, actors, decisions, routes, maps and context.
Media & Archive	Assets, ingest, transcripts, permissions, Drive folders and

Navigation item	Purpose
	derivatives.
Outputs	Articles, films, audio, photo essays, talks, courses, installations and corrections.
FCIF Library	Manual, standards, capture taxonomy, consent cards, fieldbooks and revisions.
Search	Hybrid universal retrieval and saved investigations.
Settings	Workspace, roles, integrations, taxonomies, retention, AI, exports and audit.

5.2 Universal command palette

- Create: deployment, target, interaction, interview, claim, evidence, system, place, asset, cognition update, task, output.
- Navigate: any entity by exact name, external ID, tag, city, event, or status.
- Act: start interview, open Field Mode, record field note, ingest files, verify claim, create follow-up, export source packet.
- Retrieve: natural-language query with filters and explainable result sources.
- Admin: run sync, view job failures, export workspace, and open audit event.

6. Screen-by-screen product specification

Screen specification convention

Each screen includes purpose, entry points, required components, primary actions, states, permissions, responsive behavior, and acceptance criteria. Requirement IDs use SCR-<area>-<number>.

6.1 Command Center

Route: /

Give the user a trusted answer to what matters now.

Specification area	Requirements
Required components	Global search and command button; Active deployments cards; Today timeline and route; Urgent commitments; Interview/outreach queue; Unverified high-priority claims; Assets awaiting ingest or metadata; Capture packages incomplete; Outputs in production; Sync/job health; Quick capture rail
Primary actions	Open deployment; Complete or snooze task; Start Field Mode; Create any record; Run targeted sync; Review alert
Key states	Empty first-run; Normal; Offline with cached data; Integration degraded; Sensitive item hidden by role
Mobile behavior	Cards collapse into Today, Next Interview, Capture, and Alerts. Quick Capture remains fixed at bottom.

Specification area	Requirements
Acceptance criteria	Loads actionable content in under 2 seconds on normal connection; Every alert links to its cause and resolution action; No count is displayed without queryable underlying records

6.2 Deployments index

Route: /deployments

Compare and enter all field deployments.

Specification area	Requirements
Required components	Status tabs; Calendar/timeline; Map; Deployment cards; Readiness score; Upcoming dates; Filter by city, outlet, lane and status
Primary actions	Create deployment; Duplicate structure; Archive deployment; Open planning or field workspace
Key states	Planning; Ready; Active; Processing; Closed; Cancelled
Mobile behavior	List view sorted by proximity; map optional.
Acceptance criteria	Manhattan and Reykjavík appear with dates, mission, next action and Drive status; Archived deployments remain searchable

6.3 Deployment workspace

Route: /deployments/[id]

Operate one trip from planning through outputs.

Specification area	Requirements
Required components	Mission and assignments; Readiness meter; Dates, city, timezone, credentials and Drive link; Targets; Schedule/route; Interview pipeline; Research questions; Claims/law/evidence; Shot lists and capture packages; Daily logs; Assets; Outputs; Follow-ups; Notion mirror status
Primary actions	Add target; Send/create outreach draft; Book interaction; Start Field Mode; Create daily plan; Run completeness audit; Close deployment
Key states	Planning; Field active; Post-field processing; Closed; immutable summary with continuing follow-ups
Mobile behavior	Defaults to Today tab during active dates; offline cache includes itinerary, contacts, consent and capture cards.
Acceptance criteria	A user can reach any deployment record within two taps; Readiness identifies exact missing prerequisites; Closing requires unresolved promises and restrictions review

6.4 Today / Field plan

Route: /deployments/[id]/today

Translate the deployment into a chronological, location-aware action plan.

Specification area	Requirements
Required components	Timeline; Map/route; Travel buffers; Access requirements; Interview cards; Capture package state; Evidence requests; Weather/manual notes; Emergency contacts; Offline status
Primary actions	Check in; Start capture; Record delay; Reorder; Skip with reason; Open navigation; Add spontaneous interaction
Key states	On time; At risk; Delayed; Skipped; Completed
Mobile behavior	Large tap targets; minimal text; high contrast; one-hand use.
Acceptance criteria	Changes persist offline and sync without duplicate actions; Skipped priority-A item requires reason and recovery task

6.5 Target / relationship record

Route: /people/[id] or /institutions/[id]

Preserve who the source is, why they matter, and every relationship context.

Specification area	Requirements
Required components	Identity and pronunciation; Roles and organizations over time; Expertise and interests; Contact routes; Relationship history; Deployments and interactions; Claims/evidence; Permissions/preferences; Introductions; Commitments; Potential outputs; Risk/conflict disclosures
Primary actions	Create outreach; Schedule interview; Request introduction; Add note; Link to claim/system/output; Restrict identity
Key states	Prospect; Contacted; Active source; Research participant; Collaborator; Dormant; Do not contact
Mobile behavior	Compact identity, next action, contact, and last interaction first.
Acceptance criteria	No duplicate person is created when email or external IDs match; Restricted identity is omitted from unauthorized search and exports

6.6 Interaction / interview workspace

Route: /interactions/[id]

Prepare, conduct, and close a source interaction.

Specification area	Requirements
Required components	Context and role; Participants; Purpose; Question tree; Ground rules; Consent recorder; Recording files; Live notes; Promises; Claims detected; Evidence requested; Cognition update; Follow-up
Primary actions	Begin consent; Start/stop interview timer; Mark on/off record segment; Add timestamped note; Capture claim; Close interaction

Specification area	Requirements
Key states	Draft; Scheduled; In progress; Processing; Completed; Restricted; Cancelled
Mobile behavior	Interview Mode hides unrelated app elements and prevents accidental navigation.
Acceptance criteria	Consent state is visible continuously during recording; Off-record segments are timestamped and excluded from default transcript retrieval; Closing prompts for promises, next source and immediate cognition update

6.7 Claim record

Route: /claims/[id]

Represent one exact assertion independently from its support.

Specification area	Requirements
Required components	Exact wording; Claimant/context/time; Type; Stakeholders/consequences; Evidence links; Counterevidence; Conflicts; Legal significance; Financial interest; Verification history; Permitted outputs
Primary actions	Add evidence; Request response; Change status with reason; Create contradiction; Generate verification brief
Key states	Unreviewed; Unverified; Partially verified; Verified; Disputed; False; Superseded; Do not use
Mobile behavior	Quick capture stores exact wording and context first; detailed fields later.
Acceptance criteria	Every status change is audited; Verified status requires at least one evidence record and reviewer/owner action; AI cannot set final status

6.8 Evidence / law record

Route: /evidence/[id]

Store and evaluate the source supporting or challenging claims.

Specification area	Requirements
Required components	Source identity; Evidence type; Original file/URL; Date/version/jurisdiction; Method/population/comparator/metric; Reliability/fitness; Limitations; Claims supported/challenged; Legal authority/citation; Verification notes; Drive location
Primary actions	Attach/copy to Drive; Extract text; Link claim; Add limitation; Create superseding version
Key states	Unreviewed; Authenticated; Method reviewed; Verified for stated purpose; Disputed; Superseded; Restricted
Mobile behavior	Capture URL/photo/document now; detailed evaluation can be queued.
Acceptance criteria	Source version is preserved; A later version does not overwrite the historical source; Legal records store effective dates and temporal status

6.9 Systems reconstruction

Route: /systems/[id]

Make the actual workflow, authority and consequence visible.

Specification area	Requirements
Required components	System summary; Trigger; Actors/institutions; Step graph; Inputs/outputs; Decision rules; Law; Evidence/assets; Exceptions; Feedback; Failure and repair; Affected people; Version comparison
Primary actions	Add step; Connect nodes; Mark observed/claimed; Create exception;

Specification area	Requirements
	Compare jurisdiction/time; Export map
Key states	Draft; Partially observed; Verified; Disputed; Superseded
Mobile behavior	Field capture supports adding a step or friction point with voice and photo.
Acceptance criteria	Every graph edge has a source status; Observed and claimed steps render differently; Export can include evidence citations and uncertainty legend

6.10 Media ingest

Route: /archive/ingest

Protect and convert raw field media into searchable records.

Specification area	Requirements
Required components	Device/session selector; File manifest; Duplicate detection; Destination Drive folder; Naming preview; Metadata defaults; Consent/restriction inheritance; Checksum/status; Proxy/transcript queue; Errors
Primary actions	Scan files; Assign subject; Create Drive folders; Copy/upload; Verify; Queue transcription; Quarantine
Key states	Pending; Copying; Verifying; Protected; Processing; Needs review; Failed
Mobile behavior	Mobile supports registering a file and uploading small urgent audio; large media typically ingests from desktop.
Acceptance criteria	Source files are not deleted by the application; A file is protected only after two verified copies or explicit override; Partial failure is resumable

6.11 Asset record

Route: /assets/[id]

Preserve provenance, permissions, context and potential uses for one media/document asset.

Specification area	Requirements
Required components	Preview/proxy; Master location; Capture metadata; People/place/system/claim links; Consent and restrictions; Transcript; Derivative lineage; Verification; Potential outputs; Usage history
Primary actions	Edit metadata; Create derivative; Restrict; Link records; Export source card; Request permission
Key states	Unreviewed; Protected; Metadata complete; Restricted; Approved for specified uses; Published; Do not use
Mobile behavior	Quick asset card emphasizes restriction and context; proxy-only playback.
Acceptance criteria	Original master is immutable in app; Derivatives identify parent asset and edit method; Permission changes flag affected outputs

6.12 Universal search and research canvas

Route: /search

Retrieve across structured, textual, graph, geographic and multimodal records.

Specification area	Requirements
Required components	Natural language box; Filters; Result groups; Why this matched; Source excerpts; Graph neighborhood; Map/timeline; Saved query; Research canvas
Primary actions	Open source; Pin to canvas; Compare; Create brief; Export source set; Convert question to research task
Key states	Keyword; Semantic; Hybrid; Graph neighborhood; Exact citation
Mobile behavior	Recent searches and voice query; no automatic sensitive result preview.
Acceptance criteria	Every generated answer cites internal record IDs; Search respects permissions before retrieval; User can inspect query filters and source set

6.13 Output workspace

Route: /outputs/[id]

Turn evidence into a controlled publication or media object.

Specification area	Requirements
Required components	Brief and audience; Assignment/outlet; Source set; Claims allowed; Asset selects; Structure; Draft/script; Rights and permissions; Fact check; Legal/editorial review; Versions; Publication/correction
Primary actions	Assemble source packet; Generate outline; Draft from approved sources; Request permission; Approve version; Publish/archive; Issue correction
Key states	Idea; Assigned; Researching; Drafting; Review; Ready; Published; Corrected; Killed
Mobile behavior	Mobile permits review and notes, not final publication by default.
Acceptance criteria	Generated text cannot cite restricted/off-record material; Publication readiness blocks on unresolved permissions and high-risk unverified claims; Corrections preserve prior version and reason

6.14 FCIF Library

Route: /library

Keep the method, standards, fieldbooks and capture cards inside the operational product.

Specification area	Requirements
Required components	Manual sections; Standards; Capture taxonomy; Consent cards; Domain fieldbooks; Version history; Linked implementation templates
Primary actions	Read; Search; Apply template; Propose revision; View superseded rule
Key states	Draft standard; Active; Superseded; Archived
Mobile behavior	Offline cache of active field cards and consent language.
Acceptance criteria	Every operational template identifies its governing standard version; Changing a standard creates a migration/review notice rather than silently changing historical records

6.15 Settings and integrations

Route: /settings

Administer the workspace without exposing secrets or complexity during field use.

Specification area	Requirements
Required components	Profile/workspace; Roles; Notion connection; Drive connection; AI/transcription providers; Controlled vocabularies; Retention; Exports; Audit log; Sync conflicts; Feature flags
Primary actions	Connect/revoke; Test; Run sync; Rotate secret; Export; Restore; Invite role
Key states	Healthy; Degraded; Paused; Action required
Mobile behavior	Admin screen is desktop-first.
Acceptance criteria	Secrets never render after entry; Revoking an integration stops future jobs without deleting canonical records; Sync conflict requires explicit resolution

7. End-to-end workflows

Workflow	Normative flow
WF-01 Deployment planning	Create deployment → define mission/outlets/questions → import events/targets → rank → confirm access/travel → create Drive structure → create Notion mirror → readiness review → activate.
WF-02 Source acquisition	Create/resolve target → research dossier → outreach draft → owner approval/send → response capture → schedule → prepare question tree/evidence ask → interview readiness.
WF-03 Interview	Open interaction → establish role/ground rules → record consent → capture interview and notes → mark restricted segments → close → protect files → cognition update →

Workflow	Normative flow
	claims/evidence/promises → follow-up.
WF-04 Claim verification	Create exact claim → identify claim type and stakes → attach offered evidence → retrieve independent evidence/law → assess fitness/limits/conflicts → request response → owner verification decision → permitted-language note.
WF-05 Field media ingest	Register capture session → copy to working storage → second copy → upload to Drive → verify counts/hashes/spot playback → create asset records → inherit consent → proxies/transcripts → metadata review.
WF-06 System reconstruction	Create system → define trigger/outcome → add actors and steps from observed evidence → label claimed/observed/inferred → add law and exceptions → identify affected people → verify edges → compare version/place.
WF-07 Output production	Create output → define outlet/audience/assignment → lock approved source set → outline → draft/edit → fact check → permissions check → legal/editorial review → export/publish → archive → correction monitoring.
WF-08 Notion mirror	Canonical change event → decide mirror eligibility → transform to page/database properties → upsert by external ID → store Notion ID and revision → pull permitted Notion edits/comments → conflict queue.
WF-09 Drive vault	Deployment creation → deterministic folders → source file registration → upload/link → store Drive ID → label/permissions → verify → derivatives/transcripts linked → retention and audit.
WF-10 Longitudinal revisit	Open place/system/person → review prior state → identify changes and unresolved questions → create new deployment interaction → compare claims/laws/system versions → produce change record.

7.1 Workflow state guarantees

- Every state transition is timestamped and attributable to a user or service actor.
- Irreversible actions require explicit confirmation and, where possible, soft deletion or version preservation.
- Background jobs must be idempotent: retries cannot create duplicate pages, files, transcripts, or relationships.
- An offline-created record receives a client-generated UUID and merges without losing local timestamps.
- A workflow can be incomplete without being ambiguous: missing items are represented as explicit tasks or readiness defects.
- Restricted information cannot enter a downstream workflow unless its permission policy allows the intended use.

8. Mobile Field Mode

8.1 Design goals

- One-handed operation, high contrast, large targets, minimal typing, voice-first optional capture.
- No requirement for reliable connectivity during interviews or movement.

- No accidental exposure of source identities or restricted notes on lock screen or notifications.
- Critical context—next appointment, route, contact, role, consent, evidence ask, capture package—available within two taps.
- Fast transition between Observe, Interview, Claim, Evidence, Asset, and Cognition capture.

8.2 Field Mode home

Element	Behavior
Now card	Current location/appointment, arrival buffer, contact, access instructions and one central question.
Capture progress	Minimum package grouped as Video, Photography, Audio, Records; shows missing critical items.
Consent button	Opens role-specific script and records terms with timestamp.
Quick capture	New person, interaction, claim, evidence, asset, friction, field note.
Offline indicator	Shows unsynced items and available device storage estimate.
Safety/privacy button	Instantly hides sensitive names and pauses recording metadata display.
Close subject	Runs missing-reality check and creates next actions.

8.3 Offline synchronization

Client-generated UUID + local timestamp

- > encrypted IndexedDB queue
 - > optimistic local UI
 - > connectivity detection
 - > authenticated batch sync
 - > server idempotency key check
 - > canonical insert/update
 - > relation resolution
 - > conflict result returned
 - > local queue marked synced or requires review
- Offline data must be encrypted where supported and automatically removed from local cache after successful sync according to workspace policy.
 - Large media files are not silently uploaded over cellular by default; user sets Wi-Fi-only or explicit urgent upload.
 - Consent recordings and critical interview audio receive priority upload once connectivity is restored.
 - Conflicts use field-level resolution for editable notes and append-only resolution for claims, evidence, permissions and audit events.

9. Canonical data model and Supabase schema

9.1 Modeling rules

- Every first-class entity uses UUID primary key plus human-readable stable code where operationally useful.

- Tenant-ready: every workspace-owned table includes workspace_id and RLS policies even during single-user release.
- Created_at, updated_at, created_by, updated_by, deleted_at and version are standard unless the table is immutable/event-only.
- Many-to-many relationships use explicit join tables when the relationship carries meaning, role, order, permission, or dates.
- Claims and evidence are never represented as one record type in the canonical schema, although the UI may present a combined queue.
- Permissions and restrictions are versioned policies, not free-text flags alone.
- External integrations use dedicated mapping tables and never replace canonical UUIDs.

Table	Purpose
workspaces	Tenant root, settings, retention, feature flags.
profiles	User profile and workspace membership metadata.
workspace_members	Roles and status per workspace.
deployments	Trip/event/residency mission and lifecycle.
deployment_assignments	Outlet/project-specific assignment inside deployment.
events	Conference sessions, meetings, public events and deadlines.
places	Physical/virtual places with address, geospatial and sensory fields.
routes	Ordered path between places, access and timing.
people	Canonical person identity and source profile.
organizations	Companies, agencies, universities, outlets and institutions.
person_roles	Person-organization roles over time.
relationships	User/source relationship context and stewardship.
targets	Deployment-specific pursuit reason, priority and action.
interactions	Outreach, interview, observation, meeting, panel or correspondence.
interaction_participants	People/organizations and their role in interaction.
consent_records	Recorded terms, attribution, recording and use permissions.
permission_policies	Versioned reusable policy objects attached to assets/interactions.
commitments	Promises by user or source with due date and status.
tasks	Operational actions and reminders.
research_questions	Questions, hypotheses and priority per deployment/system.
claims	Exact assertions, context, stakes and verification state.

Table	Purpose
evidence	Documents, observations, datasets, testimony and source evaluation.
claim_evidence	Support/challenge relation, strength, and notes.
laws_policies	Authority, jurisdiction, effective period and temporal status.
contradictions	Linked records that appear inconsistent; interpretation state.
absence_records	Bounded missing voice/data/question observations.
systems	Named real-world workflow or institution process.
system_steps	Ordered/graph steps, decisions, handoffs and state.
system_edges	Directed connections between steps with evidence status.
system_actors	People/organizations and authority/experience roles.
observations	Structured field observations and scope limits.
cognition_updates	Pre/post hypothesis, evidence causing change and uncertainty.
capture_packages	Required and stretch capture sets for subject.
capture_items	Individual required item and completion/evidence.
assets	Media/document/object record and master/proxy metadata.
asset_relations	Asset links to any supported entity with relation type.
asset_derivatives	Parent-child lineage and transformation disclosure.
transcripts	Versioned transcript and speaker/timestamp data.
outputs	Article, film, audio, map, course, talk, installation or brief.
output_sources	Claims/evidence/assets/people used by output.
corrections	Published correction and affected claims/assets.
tags	Controlled and user-created tags.
entity_tags	Tag relation.
external_mappings	Notion/Drive/calendar/email/provider IDs and sync versions.
sync_jobs	Integration job state, idempotency and error.
ai_runs	Prompt, model, inputs, outputs, review and cost metadata.
embeddings	Vector chunks linked to authorized source records.

Table	Purpose
audit_events	Append-only actor/action/before-after/security record.
notifications	User-facing alerts and resolution state.
saved_searches	Persisted structured/hybrid queries and sharing policy.

9.2 Selected table specifications

9.2.1 deployments

- id uuid pk
- workspace_id uuid fk
- code text unique per workspace
- name text
- deployment_type enum
- status enum
- start_at/end_at timestamptz
- timezone text
- city/country text
- mission text
- primary_question text
- priority enum
- drive_root_external_id text
- notion_page_external_id text
- readiness_score numeric
- created/updated/version/audit fields

9.2.2 people

- id
- workspace_id
- preferred_name
- legal_or_full_name encrypted/nullable
- pronunciation
- email/phone encrypted
- bio
- expertise text[]
- relationship_status
- sensitivity_level
- do_not_contact boolean
- merge_parent_id
- timestamps/version

9.2.3 interactions

- id
- workspace_id
- deployment_id
- interaction_type
- status

- title
- scheduled_at/start/end
- timezone
- location_id
- purpose
- capacity enum editorial/research/networking/commercial/mixed
- attribution_default
- recording_status
- summary
- restricted_notes encrypted
- immediate_follow_up_due
- timestamps/version

9.2.4 consent_records

- id
- workspace_id
- interaction_id
- subject_person_id
- captured_at
- capture_method
- audio_asset_id
- recording_allowed
- photo_allowed
- video_allowed
- transcription_allowed
- publication_allowed
- research_allowed
- commercial_contact_allowed
- attribution_mode
- allowed_descriptor
- embargo_until
- restrictions text
- withdrawal_terms text
- supersedes_consent_id
- created_by

9.2.5 claims

- id
- workspace_id
- deployment_id
- interaction_id
- claimant_person_id/org_id
- exact_text
- claim_type
- context
- made_at
- verification_status
- confidence_user
- stakes
- financial_interest
- legal_significance
- publication_language

- supersedes_claim_id
- timestamps/version

9.2.6 evidence

- id
- workspace_id
- evidence_type
- title
- source_person/org
- source_url
- drive_file_id
- publication_date
- version_label
- jurisdiction
- methodology
- population
- comparator
- metrics
- limitations
- reliability_status
- fitness_for_purpose
- authenticated_at/by
- sensitivity_level
- timestamps/version

9.2.7 assets

- id
- workspace_id
- asset_code
- asset_type
- capture_type_id
- master_provider
- master_external_id
- master_path
- original_filename
- canonical_filename
- mime_type
- bytes
- captured_at
- timezone
- place_id
- capture_device
- hash_sha256
- ingest_status
- permission_policy_id
- sensitivity_level
- transcript_status
- proxy_storage_path
- immutable boolean
- timestamps/version

9.2.8 outputs

- id
- workspace_id
- deployment_id
- output_type
- title
- status
- outlet
- audience
- assignment
- deadline
- risk_level
- source_lock_at
- permissions_status
- fact_check_status
- review_status
- published_at
- publication_url
- drive_archive_id
- current_version
- timestamps

9.2.9 external_mappings

- id
- workspace_id
- entity_type
- entity_id
- provider enum
- external_id
- external_url
- external_parent_id
- last_pushed_version
- last_pulled_version
- last_synced_at
- sync_direction
- sync_status
- conflict_payload jsonb

9.3 Relationship rules

Relationship	Cardinality	Rule
Deployment → target	one-to-many	A target is contextual to a deployment even when the same person appears in other deployments.
Person ↔ organization	many-to-many through person_roles	Preserves title, dates, jurisdiction, verification and source.
Interaction ↔ participants	many-to-many	Stores participant capacity, attendance, attribution and source role.
Claim ↔ evidence	many-to-many through claim_evidence	Relation states

Relationship	Cardinality	Rule
		supports/challenges/contextualizes and evidentiary strength.
System ↔ steps	one-to-many plus graph edges	Ordered and branching workflows supported.
Asset ↔ entities	polymorphic relation table	One asset may depict a person, evidence a claim, document a step and belong to a deployment.
Output ↔ source records	many-to-many through output_sources	Locks the exact source set and permitted use at each version.
Permission policy → assets/interaction	one-to-many/versioned	Restrictions propagate through source relationships and derivative lineage.
External mapping → any canonical entity	one-to-one per provider/entity	Allows idempotent Notion and Drive sync.

10. API and integration specification

10.1 API conventions

- Base path /api/v1; JSON over HTTPS; generated OpenAPI document is normative for request/response shape.
- Authentication uses Supabase JWT. Server verifies workspace membership and RLS plus service-level authorization.
- Every write supports Idempotency-Key header for mobile/offline and integration retries.
- List endpoints use cursor pagination, filter objects, sort, sparse fieldsets, and include options.
- Errors use RFC 9457-style problem details: type, title, status, detail, instance, code, field_errors, retryable.
- All mutation responses include entity id, version, updated_at and audit_event_id.
- ETag or version precondition is required for conflict-sensitive updates.
- No endpoint returns restricted transcript or identity fields unless authorization and intended-use check pass.

Method	Endpoint	Purpose
POST	/deployments	Create deployment and optionally provision Drive/Notion asynchronously.
GET	/deployments	Filter/list deployments.
GET/PATCH	/deployments/{id}	Read/update deployment.
POST	/deployments/{id}/activate	Validate readiness and enter field-active state.
POST	/deployments/{id}/close	Run closure gates and close.
POST	/targets	Create deployment target.
POST	/targets/{id}/outreach-draft	Generate grounded outreach draft; never send.
POST	/interactions	Create/schedule interaction.

Method	Endpoint	Purpose
POST	/interactions/{id}/consent	Record consent policy and optional audio asset.
POST	/interactions/{id}/close	Close interview and create follow-up prompts.
POST	/claims	Quick or full claim capture.
POST	/claims/{id}/evidence-links	Link evidence with relation.
POST	/claims/{id}/verify	Human-only verification transition.
POST	/evidence	Create evidence or law record.
POST	/systems	Create system.
POST	/systems/{id}/steps	Add system step.
POST	/systems/{id}/export	Generate diagram/source packet.
POST	/assets/register	Register media before or during ingest.
POST	/assets/{id}/protect	Confirm backup/Drive protection state.
POST	/assets/{id}/transcribe	Queue transcript.
POST	/ingest/sessions	Create ingest session/file manifest.
POST	/ingest/sessions/{id}/verify	Verify manifest and copies.
POST	/search	Hybrid authorized search.
POST	/research/brief	Generate cited internal brief from selected records.
POST	/outputs	Create output workspace.
POST	/outputs/{id}/source-lock	Lock sources and permissions.
POST	/outputs/{id}/readiness	Evaluate publication readiness.
POST	/sync/notion	Run or schedule Notion sync.
POST	/sync/drive	Run or schedule Drive sync.
GET	/sync/jobs/{id}	Inspect job state/errors.
POST	/exports/workspace	Create portable export package.

10.2 Webhook/event endpoints

- POST /webhooks/notion: validate signature/provider token; enqueue page/data-source change processing.

- POST /webhooks/drive: receive/acknowledge change notifications; fetch canonical change details before mutation.
- POST /webhooks/transcription: receive job completion; verify provider request; store result as new transcript version.
- POST /webhooks/vercel is not part of application domain; deployment events remain in CI/observability.
- Webhook handlers acknowledge quickly, persist raw event, deduplicate by provider event ID, and process asynchronously.

11. Notion synchronization

11.1 Purpose and boundary

Notion is a curated knowledge mirror and fail-safe interface. It is not the canonical database. The app must be useful when Notion is unavailable, and Notion must remain readable when the app is unavailable.

11.2 Mirrored objects

Canonical object	Notion representation	Default direction
Deployment	Page plus linked database views	FIOS → Notion; selected notes/comments may return
Person/institution	Database page with biography, relationship history and approved links	Bidirectional limited fields
Interaction	Page for preparation, notes, ground rules and follow-up	FIOS → Notion; human notes pull allowed
Claim/evidence	Databases or linked views	FIOS → Notion; status edits require conflict review
System	Page with steps and exported diagram	FIOS → Notion
Output	Project page with source and readiness summary	Bidirectional notes; canonical status in FIOS
FCIF standard/manual	Page/library	Authoritative document may originate in Drive/Git; Notion is readable mirror

11.3 Sync mechanics

- Use a dedicated Notion connection and store page/data-source IDs in external_mappings.
- Upsert by canonical entity UUID exposed as a hidden or visible External ID property.
- Store last pushed canonical version and last pulled Notion revision; detect concurrent edits.
- Whitelisted properties may be bidirectional: human notes, next action, selected dates, contact route. Claims, consent, permissions, verification, legal status and audit are canonical-only unless resolved in a conflict screen.
- Notion page body is generated in stable sections with marker comments/blocks to preserve user-authored notes outside managed regions where technically feasible.
- Deleted canonical entities are archived in Notion, not hard-deleted automatically.
- Notion API rate limits and pagination require queued, incremental sync rather than full rewrites.

11.4 Conflict policy

Conflict type	Resolution
Notion note changed; canonical note unchanged	Import as a new interaction/note version and preserve author/time.
Canonical and Notion editable field both changed	Create conflict record; user chooses canonical, Notion, or merged value.
Notion changes protected field	Reject automatic import; show proposed change and rationale.
Notion page missing	Recreate or mark mirror disabled; never delete canonical record.
Mapping points to duplicate Notion pages	Select survivor, archive duplicate, preserve URLs in audit.

12. Google Drive media-vault synchronization

12.1 Vault principles

- Drive stores original masters and source-document copies because it is independently accessible and exportable.
- Application access should be scoped to a dedicated Field Intelligence OS folder or shared drive where feasible, rather than broad access to all personal Drive content.
- Drive folder/file IDs—not paths alone—are canonical external references because names and parent folders may change.
- Application-generated metadata never substitutes for embedded capture metadata or source provenance; both are retained.
- The application does not delete original masters automatically.

12.2 Deterministic folder template

Field Intelligence OS/
 00 Canon and Standards/
 01 Deployments/
 YYYY-MM City/
 00 Administration/
 01 Plans and Credentials/
 02 Consent and Permissions/
 03 Interviews/
 04 Photography Masters/
 05 Video Masters/
 06 Audio Masters/
 07 Documents and Evidence/
 08 Routes and Spatial Data/
 09 Transcripts and Proxies/
 10 Outputs/
 11 Exports and Closeout/
 02 People and Institutions/
 03 Systems and Places/
 04 Published Outputs/
 05 Archive Administration/

12.3 File operations

Operation	Requirement
Provision	Deployment creation can enqueue folder creation and store each folder ID.
Register	Create asset/file record before upload when possible; use session manifest for bulk ingest.
Upload/copy	Large files may use resumable upload; existing Drive files may be linked without duplicate upload.
Rename	Apply canonical filename after verified copy; preserve original filename in asset record.
Verify	Record size, MIME, duration/dimensions where available, and checksum if locally generated.
Labels	Optional Drive labels may carry deployment, sensitivity and record ID; database remains canonical.
Permissions	Default private; share only through explicit workflow; source restrictions may prohibit broader sharing.
Changes	Drive change notifications enqueue metadata reconciliation; never treat filename changes as a new asset.
Delete	App defaults to archive/quarantine request; permanent deletion requires owner confirmation and audit.

13. AI and retrieval architecture

13.1 AI feature classes

Feature	Permitted function
Transcription	Convert audio/video to timecoded text with speaker segmentation; human correction required for names, technical and legal language.
Extraction	Propose people, organizations, claims, evidence references, laws, promises, questions, system steps and permissions from source material.
Classification	Suggest capture type, sensitivity, project lanes, tags, claim type, and likely output.
Retrieval	Hybrid keyword, filters, vectors and graph-neighborhood search with source-level permission filtering.
Comparison	Compare jurisdictions, versions, claims, systems, places, interviews and longitudinal changes.

Feature	Permitted function
Planning	Propose target rankings, question trees, missing coverage and next actions from existing records.
Synthesis	Create cited briefs, timelines, evidence matrices, interview packets, system maps and output outlines.
Production assistance	Suggest selects, transcript-based assemblies, captions, derivatives and source cards; no fabricated documentary content.
Quality control	Identify unsupported statements, permission conflicts, missing perspectives, stale law, contradictions and duplicate records.

13.2 AI non-delegable decisions

- Final consent interpretation, attribution, off-record boundaries, privacy classification, or source identity disclosure.
- Final verification or falsity decision for a claim.
- Legal advice, regulatory conclusion, or determination that law is current without primary-source review.
- Final publication, outreach sending, correction issuance, or destruction of records.
- Changing a quotation, creating an interview response, fabricating ambience, or presenting synthetic media as field evidence.
- Deciding that a human participant had capacity to consent.

13.3 Retrieval pipeline

Query + user intent + intended use

- > workspace and role authorization
- > structured filters (deployment, date, entity, jurisdiction, permission)
- > keyword/full-text candidates
- > vector candidates from authorized chunks
- > graph-neighborhood candidates
- > merge + rerank
- > source diversity and recency checks
- > result excerpts with record IDs and permission labels
- > optional cited synthesis
- > user inspection / save to research canvas

13.4 Embedding/chunk policy

- Chunk transcripts by speaker turns and topical windows while retaining timecodes and interaction ID.
- Chunk documents by headings/pages/sections and retain version, page, source URL and evidence ID.
- Do not embed off-record or highly restricted material into broadly accessible indexes; use segregated indexes or exclude entirely.
- Embedding deletion or permission change must remove or re-scope all derived chunks.
- Generated summaries are not embedded as substitutes for originals unless labeled derived and linked to source set.

13.5 AI run record

- Model/provider and version
- Feature and prompt/template version

- User and intended use
- Input record IDs and content hashes
- Permission scope
- Output and confidence/proposed fields
- Human review status and edits
- Token/cost/latency
- Safety or policy flags
- Created time and retention

14. Automations and event processing

Trigger	Automation
Deployment created	Provision Drive folders; create Notion mirror; seed standard capture packages; create readiness tasks.
Interview scheduled	Generate preparation checklist; fetch target dossier; create consent prompt; cache Field Mode card.
Interaction closed	Prompt cognition update; extract proposed claims/promises; queue follow-up; protect media.
Asset protected	Queue proxy, metadata extraction, transcription, thumbnail and embedding according to permission.
Claim created	Search existing evidence and contradictions; create verification task; flag high-risk law/health/financial claims.
Law/policy record aging	Create re-verification alert based on volatility and publication/deployment date.
Permission changes	Recalculate asset/output eligibility; block affected publication; notify owner.
Output source lock	Snapshot source versions, permissions and verification status.
Deployment active date	Switch app to Field Mode by default and cache deployment package offline.
Daily field close	Run coverage audit; list unsynced files; unresolved promises; missing minimum packages; tomorrow plan.
Notion/Drive webhook	Persist event, deduplicate, queue scoped sync and surface conflicts.
Job failure	Retry with backoff; after threshold, alert with human resolution instructions.

14.1 Job model

- Job types, payload schema version, idempotency key, priority, run_after, attempts, max_attempts, status, lock owner/time, result, error and linked entity.
- Jobs processing restricted data must execute only in approved providers/regions and record data-handling policy.
- Long jobs must expose progress and be cancelable when safe.

- Automations create proposed records or tasks when interpretation is uncertain; they do not silently finalize substantive conclusions.

15. Permissions, security, privacy, and editorial boundaries

15.1 Permission model

Layer	Controls
Workspace membership	Owner, investigator, editor, researcher, producer, viewer, service account.
Entity sensitivity	Public; internal; confidential; source-protected; highly restricted.
Intended use	Editorial, research, professional networking, commercial, legal/privileged, private cognition.
Action permission	View, create, edit, link, export, share, publish, administer.
Field permission	Certain identity, contact, health, consent, off-record and privileged fields are separately protected.
Output permission	Source and asset policy evaluated against output type, audience, outlet and publication state.
Integration permission	Provider scopes and per-entity mirror/export eligibility.

15.2 Role matrix summary

Resource	Owner	Investigator	Editor	Researcher	Producer	Viewer
Workspace settings	A	-	-	-	-	-
Deployments/tasks	A	E	E	E	R	R
People/relationships	A	E	E	E	R	R
Restricted identities	A	R	R	-	-	-
Interactions	A	E	E	E	R	R
Consent/permissions	A	E	R	-	R	-
Claims/evidence	A	E	E	E	R	R
Verification decision	A	E	E	R	-	-
Assets/ingest	A	E	R	-	E	-

Resource	Owner	Investigator	Editor	Researcher	Producer	Viewer
Outputs/drafts	A	E	E	R	R	R
Publish/export	A	-	E	-	-	-
Audit log	A	E	E	E	R	R

Legend: A = administer; E = edit/act; R = read within sensitivity scope; - = denied by default.

15.3 Security requirements

ID	Requirement
SEC-01	All traffic encrypted; secrets stored only in managed environment-secret systems.
SEC-02	Supabase RLS enabled on all workspace tables before client access.
SEC-03	Sensitive contact, private cognition, health and restricted notes encrypted at application or database column level where practical.
SEC-04	Audit logs append-only for consent, permissions, verification, export, deletion and publication actions.
SEC-05	MFA required for owner and any role with restricted-source or publication access.
SEC-06	Session timeout and device revocation available; mobile app supports biometric/PIN re-entry if feasible.
SEC-07	No sensitive content in notification previews, analytics payloads, client logs or error trackers.
SEC-08	Drive and Notion tokens encrypted, least-privilege, revocable and separated by environment.
SEC-09	Exports are watermarked/logged when containing restricted content and expire where shared by temporary link.
SEC-10	Backups and restore procedures tested; Supabase database backups do not substitute for Drive media backup.
SEC-11	Dependency, secret and migration scanning in CI.
SEC-12	Incident response includes source-protection triage and permission-impact analysis.

15.4 Editorial/research/commercial separation

- Every interaction has a capacity field; mixed capacity requires a role-transition event.
- Editorial access does not create research consent or commercial permission.
- Research participation does not create public attribution or identifiable multimedia permission.

- Commercial opportunities discovered through source work are placed in a separate queue and require express follow-up permission.
- AI retrieval filters by intended use so material collected under one boundary is not casually reused in another.
- Output readiness evaluates both legal permission and ethical context; “technically public” is not automatically appropriate.

16. Offline, media, accessibility, and performance requirements

ID	Requirement
NFR-01 Availability	Core canonical app target 99.5% monthly after beta; field offline mode mitigates connectivity.
NFR-02 Performance	Command Center p75 LCP <2.5s on broadband; common authenticated API p95 <500ms excluding external providers.
NFR-03 Search	Structured search p95 <1s; hybrid synthesis begins streaming within 3s and exposes progress.
NFR-04 Mobile	Core capture actions usable at 360px width; no horizontal scrolling for primary flows.
NFR-05 Accessibility	WCAG 2.2 AA target; keyboard navigation, focus states, labels, captions/transcripts, reduced motion and high contrast.
NFR-06 Media	Proxy playback adaptive; original masters never required to load list pages.
NFR-07 Offline	Today plan, consent cards, target briefs, capture package and unsynced drafts available offline.
NFR-08 Internationalization	Unicode-safe; local timezone, date, names and pronunciation; architecture ready for multiple languages.
NFR-09 Exportability	Workspace can export structured JSON/CSV plus linked file manifest and readable HTML/PDF package.
NFR-10 Durability	Schema migrations reversible or forward-fix documented; append-only records preserved.

16.1 Media strategy

- Original master: Drive only by default, immutable from app.
- Proxy: app-optimized low-resolution media in Supabase Storage or signed Drive stream.
- Thumbnail/waveform: generated for browsing without revealing restricted content to unauthorized roles.
- Transcript: versioned text/JSON with timecodes and speaker labels; restriction segments encoded.
- Derivative: export file linked to parent master, edit project, AI involvement, captions, and output.
- Integrity: optional local SHA-256 at ingest; at minimum file size and metadata verified against manifest.

17. Analytics, observability, QA, and acceptance criteria

17.1 Product analytics

- Operational analytics only by default: workflow completion, time to protect media, claims verified, follow-ups overdue, search success, sync failures, offline queue size.
- No third-party behavioral session replay on restricted screens without explicit privacy review.
- Analytics events use entity types and pseudonymous IDs, not source names or content.
- User can disable non-essential analytics.

17.2 Observability

Signal	Required coverage
Application errors	Client and server errors, release, route, workspace pseudonym, request ID; redact content.
API metrics	Latency, status, endpoint, rate limit, database time.
Background jobs	Queue age, attempts, provider, duration, failure class and linked entity.
Sync	Items scanned/created/updated/conflicted/failed, cursor and provider latency.
AI	Feature, provider/model, latency, cost, review outcome; no raw restricted prompt in general logs.
Security	Auth failures, role changes, exports, restricted views, permission changes and token revocation.
Media ingest	Manifest counts, bytes, verification, failures and protection time.

17.3 Test strategy

Layer	Coverage
Unit	Validation, permission evaluation, state machines, naming, transformation and extraction parsers.
Database	RLS policy tests, migrations, constraints, temporal/version behavior and idempotency.
Integration	Notion sandbox, Drive test vault, transcription callbacks, AI provider mocks and actual smoke tests.
E2E	Manhattan/Datamaran and Reykjavík/IMA scenarios across desktop and mobile.
Offline	Create/update/conflict and audio-note queue under loss/recovery of connectivity.
Security	OWASP checks, secret scanning, privilege escalation, IDOR, export and signed URL tests.

Layer	Coverage
Accessibility	Automated checks plus keyboard/screen-reader manual testing of core paths.
Disaster recovery	Database restore, mapping rebuild, Notion mirror regeneration and Drive manifest recovery.

17.4 v1.0 release gates

22. All core screens meet acceptance criteria and role restrictions.
23. Manhattan and Reykjavík seed data load without manual database editing.
24. End-to-end IMA and Datamaran simulations complete with audit trail.
25. Offline capture sync passes duplicate and conflict tests.
26. Notion and Drive integrations complete idempotent upsert and recover from failed job.
27. No high-severity security finding; RLS policy suite passes.
28. Workspace export and restore manifest tested.
29. Owner completes a live rehearsal and signs product acceptance record.

18. Migration and implementation plan

18.1 Source artifact migration

Source	Target
Conference Intelligence workbook: Dashboard/September Decision	Deployments, readiness configuration, decisions and tasks.
DDW/AI Expo/MJBiz/Manhattan/Reykjavík target sheets	Targets, organizations, people, events, research questions and evidence requests.
Research Sheets	Claims, evidence, laws/policies, systems and Hawai'i transfer notes.
Interview Outreach	Interactions/outreach drafts and target next actions.
Master Tracker	Tasks, commitments, statuses and follow-up dates.
Protocols/Consent Cards	FCIF standards, permission policy templates and interaction scripts.
Capture Taxonomy	capture_types reference table and Field Mode templates.
Field Capture Matrix/Shot Lists	capture_packages, capture_items and subject readiness.
Asset Metadata	assets import template.
Systems Reconstruction	systems, steps, edges and source links.
Daily Coverage Check	daily logs, missing-coverage tasks and cognition updates.
Field Investigation Manual	FCIF Library document and searchable chunks; not flattened into operational records indiscriminately.

Source	Target
OS v0.1 packet	Architecture decision baseline and migration traceability.
Existing Notion/Drive objects	external mappings and canonical seed records; preserve current URLs/IDs.

18.2 Import pipeline

Freeze source version

- > export workbook/manual to machine-readable staging
- > validate headers and controlled values
- > assign canonical UUIDs and stable codes
- > deduplicate people/organizations/places
- > load reference tables
- > load deployments/events/targets
- > load interactions/tasks/questions
- > load claims/evidence/laws/systems
- > load capture packages and output ideas
- > attach external Notion/Drive mappings
- > reconciliation report
- > owner approval
- > archive source files as migration evidence

18.3 Build phases

Phase	Deliverables
Phase 0: foundation	Repository, environments, design system, auth, workspace, schema, RLS, audit, CI/CD.
Phase 1: operating core	Command Center, deployments, targets, tasks, interactions, claims/evidence, system records.
Phase 2: field and archive	PWA Field Mode, consent, capture packages, offline queue, ingest, assets, Drive integration.
Phase 3: knowledge mirror	Notion synchronization, FCIF library, migration/import, conflict center.
Phase 4: AI/retrieval	Transcription, extraction proposals, embeddings, hybrid search, cited briefs.
Phase 5: outputs and hardening	Output workspace, permissions gates, corrections, export/restore, accessibility/security/performance.
Phase 6: live deployment	Manhattan/Reykjavík rehearsal, bug fixes, production acceptance.

18.4 Suggested engineering milestones

Milestone	Definition of done
M1 Skeleton	Authenticated app deployed; schema/migrations and RLS; navigation and design system.
M2 Planning usable	Manhattan/Reykjavík can be planned with targets, tasks, schedule and readiness.
M3 Interview usable	Consent, interaction, notes, claims, promises and follow-up complete.
M4 Field usable	Offline Today and quick capture; sync and missing-reality closeout.
M5 Archive usable	Drive provisioning, manifest ingest, asset metadata, transcription/proxies.
M6 Intelligence usable	System reconstruction, hybrid search, evidence verification and cited brief.
M7 Output usable	Source lock, draft assembly, permissions/readiness, archive and corrections.
M8 Production	Notion mirror, export/restore, security/accessibility gates, live rehearsal.

19. Manhattan and Reykjavík live seed specification

19.1 Manhattan deployment

Field	Seed value
Code	DEP-2026-NYC-SEP
Dates	20–22 September 2026
Mission	48 Hours Inside the Systems Trying to Redesign the Future
Primary lanes	Climate systems; AI/ESG/legal risk; legal tech; fintech; creator economy; clean energy; materials; JBD/NSAG; Portfolio Professional; Cannes/Hawai'i
Priority subjects	Climate Week ecosystem; Regenerative Systems Summit; Datamaran; PwC; clean-energy transition; materials; creator economy; legal tech; fintech
Primary system exemplars	ESG risk intelligence workflow; climate data to capital-project decision; creator production institution; legal-AI judgment workflow; financial-AI decision consequence
Drive mapping	Existing 2026-09 Manhattan deployment folder

Field	Seed value
Initial status	Planning / access and side meetings required

19.2 Reykjavík deployment

Field	Seed value
Code	DEP-2026-REK-SEP
Dates	20–26 September 2026
Mission	A Small Island Decides What Plant Medicine Means
Primary outlets	Fat Nugs Magazine; Cannabis Law Report
Secondary lanes	NSAG; JBD; science communication; Portfolio Professional; Cannes; Hawai'i island comparison
Priority subjects	Health for the Future; Dr. Magnús Þórsson; Icelandic Medicines Agency; Icelandic Hemp Association; University of Iceland wastewater researchers; Helgi Gunnlaugsson; Directorate of Health; Landspítali; treatment/harm reduction; journalists; product regulators; wellness/public spaces
Primary system exemplar	Legal cannabinoid medicine access pathway
Drive mapping	Existing 2026-09 Reykjavík deployment folder
Initial status	Planning / interview and institutional access requests required

19.3 Acceptance simulation A: Icelandic Medicines Agency

30. Open Reykjavík deployment and target record; see mission, outlet boundaries, questions, law and evidence asks.
31. Create/schedule regulator interaction and automatically cache consent and capture package.
32. In Field Mode, capture exterior, entrance, interview consent, exact legal explanation, guidance images, classification process and cognition update.
33. Ingest files to Drive and register assets; transcript creates proposed claims, laws and system steps.
34. Human verifies product-classification and authorization claims against primary source; link to cannabinoid access system.
35. Create “One Plant, Several Regulators” output; source lock excludes any restricted material; readiness identifies missing business/consumer perspective.
36. Notion mirror shows readable interview and legal summary; Drive retains originals; audit records every permission and status change.

19.4 Acceptance simulation B: Datamaran

37. Open Manhattan deployment and Datamaran target; review purpose, AI/ESG/legal-risk questions and requested methodology.

38. Conduct demo under screen/data permission; capture workflow and human review point.
39. Create claims about issue detection/materiality and separate evidence records for methodology/case examples.
40. Reconstruct source signal → classification → risk/materiality → legal/executive action → monitoring, with claimed/observed edges distinguished.
41. Create contradiction if “objective prioritization” conflicts with human-curated weighting; request response.
42. Produce cited internal brief and short-film source packet while enforcing corporate confidentiality and no-client-data restrictions.

20. Roadmap, risks, and open decisions

20.1 Post-v1 roadmap

Release	Direction
v1.1	Calendar/Gmail workflow, enhanced maps, batch mobile upload, more complete Notion views, external collaborator review.
v1.2	Longitudinal place/system comparison, multimedia timeline, richer transcript editor, output rights ledger.
v2.0	Team workspaces, granular project/client walls, assignment commissioning, collaborative verification and editor workflows.
v2.x	AI agent orchestration, investigation trees, automated evidence monitoring, multilingual field operations.
v3.0	Academy/certification, standards conformance, organizational deployment, APIs/SDK and open interoperability specification.

20.2 Principal risks and mitigations

Risk	Mitigation
Overbuilding before field validation	Ship vertical slices around Manhattan/Reykjavík; defer speculative modules behind feature flags.
Metadata burden	Progressive capture, defaults, AI suggestions and post-field review queues; never require every field during live work.
Three-system divergence	Supabase canonical; deterministic sync; conflict center; Notion/Drive treated as mirrors/vault.
Sensitive-source exposure	RLS, field encryption, segregated embeddings, safe notifications and explicit intended-use filters.
AI hallucination or authority drift	Proposal-only outputs, mandatory citations, human status transitions, run audit and source inspection.
Large-media cost/performance	Drive masters, lightweight proxies, resumable jobs, no master

Risk	Mitigation
	loading in lists.
External API change/limits	Adapter layer, job retries, versioned integrations and manual fallback/export.
Law becomes stale	Temporal legal model, volatility scores, scheduled reverification and publication-time checks.
Offline conflicts	Client UUIDs, idempotency, append-only protected records and visible conflict resolution.
Personal system becomes unteachable	Canonical ontology, standards/versioning, clear roles and acceptance tests from the beginning.

20.3 Open architecture decisions

ID	Decision	Resolution rule
ADR-001	Use Supabase Storage or signed Drive streaming for proxies by default?	Choose after prototype bandwidth and permission tests.
ADR-002	Queue implementation: Postgres jobs/cron, Supabase queue capability, or external managed queue?	Choose based on transcription and sync volume.
ADR-003	Encryption strategy for restricted text fields and searchable content.	Balance field encryption with authorized search.
ADR-004	Notion body merge strategy and preservation of user-written blocks.	Prototype managed-region markers and conflict behavior.
ADR-005	Map provider and offline map requirements.	Select based on cost, privacy, geocoding and downloadable route support.
ADR-006	AI provider routing and local/on-device options for restricted interviews.	Policy and cost evaluation required.
ADR-007	Whether original media should later migrate from Drive to dedicated object storage with Drive mirror.	Defer until scale or API performance requires.

Appendix A. Controlled vocabularies

Vocabulary	Initial values
deployment_status	planning, ready, active, processing, closed, cancelled
priority	A, B, C
interaction_capacity	editorial, research, professional_networking, commercial, mixed, private

Vocabulary	Initial values
attribution_mode	on_record, background_defined, anonymous, off_record_segment, confidential, undecided
verification_status	unreviewed, unverified, partially_verified, verified, disputed, false, superseded, do_not_use
evidence_type	direct_observation, primary_document, dataset, peer_reviewed_research, official_record, testimony, demonstration, media_report, expert_analysis, physical_artifact
asset_type	photo, video, audio, document, object, gps, screen_capture, note, consent
sensitivity_level	public, internal, confidential, source_protected, highly_restricted
output_status	idea, assigned, researching, drafting, review, ready, published, corrected, killed
sync_status	never, queued, syncing, synced, conflict, failed, paused

Appendix B. Domain event catalog

Event	Minimum payload
DeploymentCreated	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
DeploymentActivated	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
TargetPrioritized	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
OutreachDraftCreated	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
InteractionScheduled	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
ConsentRecorded	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
InteractionStarted	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
InteractionClosed	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
CommitmentCreated	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
ClaimCaptured	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata

Event	Minimum payload
EvidenceLinked	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
VerificationStatusChanged	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
SystemStepAdded	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
CognitionUpdated	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
AssetRegistered	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
AssetProtected	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
TranscriptCompleted	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
PermissionChanged	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
OutputSourceLocked	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
OutputPublished	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
CorrectionIssued	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
NotionSyncRequested	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
DriveSyncRequested	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
SyncConflictDetected	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata
ExportCompleted	event_id, workspace_id, actor_id/service, occurred_at, entity_id, entity_version, correlation_id, permitted metadata

Appendix C. Example API payloads

```
{
  "deployment": {
    "name": "Reykjav\u00edd Reporting Residency",
    "code": "DEP-2026-REK-SEP",
    "start_at": "2026-09-20",
    "end_at": "2026-09-26",
    "timezone": "Atlantic/Reykjavik",
```

```

"mission": "A Small Island Decides What Plant Medicine Means",
"primary_outlets": [
  "Fat Nugs",
  "Cannabis Law Report"
],
"provision_integrations": {
  "drive": true,
  "notion": true
}
}
{
  "claim": {
    "exact_text": "Sativex is the only cannabis-containing medicinal product with marketing authorization in Iceland.",
    "claim_type": "legal_regulatory",
    "claimant_organization_id": "org_ima",
    "context": "Regulator interview / public guidance",
    "verification_status": "unverified",
    "intended_outputs": [
      "Fat Nugs feature",
      "Cannabis Law Report explainer"
    ]
  }
}

```

Appendix D. Source and platform notes

Source	Design implication	URL
[S1] Notion API official reference	Connections access pages, data sources and users through a REST API; objects use UUIDs and cursor pagination. Product design should treat Notion as an integration with versioned IDs, pagination and installation tokens.	https://developers.notion.com/reference/intro
[S2] Google Drive API official overview	Drive API supports application storage integration, dedicated app folders, Picker integration, file metadata, labels and change/event workflows. FIOS should scope access to its dedicated vault where feasible.	https://developers.google.com/workspace/drive/api/guides/about-sdk
[S3] Supabase official database overview	Each project uses full Postgres; Auth, Storage, Realtime and Edge Functions build around it. Row Level Security is required before direct client access; pgvector and PostGIS support retrieval and geography.	https://supabase.com/docs/guides/database/overview
[S4] Vercel official deployment overview	Git-connected commits and pull requests create deployments with unique preview URLs; local, preview and production environments support the recommended release workflow.	https://vercel.com/docs/deployments/overview
[S5] Vercel environments	Vercel provides Local, Preview and Production, with custom environments available on some plans; each environment can use distinct variables.	https://vercel.com/docs/deployments/environments

Appendix E. Definition of ready for engineering

- Product owner approves v1.0 scope, source-of-truth rules, role model and the two acceptance simulations.
- Existing Notion database IDs and Drive folder IDs are recorded in secure environment configuration or migration seed.
- GitHub repository and Vercel project are selected; environments and secrets policy exist.
- Supabase project is created with region and backup policy approved.
- Architecture decisions ADR-001 through ADR-006 are assigned owners and target dates.
- Design prototype covers Command Center, Deployment Workspace, Interview Mode, Claim/Evidence, Field Mode and Ingest.
- Migration export of current workbook/manual is frozen and checksummed.
- Engineering backlog is generated from requirement IDs and release gates.